

A Machine Learning Approach for Intrusion Detection System on NSL-KDD Dataset

Iram Abrar, Zahrah Ayub, and Faheem Masoodi*
Department of Computer Science,
University of Kashmir, Jammu & Kashmir, INDIA
[*masoodifahim@uok.edu.in](mailto:masoodifahim@uok.edu.in)

Alwi M Bamhdi
Department of computer Sciences,
Umm Al-Qura University, Saudi Arabia

Abstract—In the field of Network Security, there is a constant expedition towards the cyber-attacks which can lead to a destabilized network. Moreover, with unexpected inception and increased use of the Internet, malicious activities in the network are at a rapid upsurge. To detect anomalies in the network, it is crucial to construct a robust intrusion detection system (IDS) that combats unauthorized access to network resources so that information can be secured. Lately, various noteworthy methods were proposed to offer a panacea to intrusion detection, yet it remains a challenge to build a secure system as the attackers subsequently change their behavior in order to dodge the security mechanism of the system. In current research work, machine learning (ML) classifiers, namely support vector machine (SVM), K-nearest neighbor (KNN), logistic regression (LR), Naïve Bayes (NB), Multi-layer Perceptron (MLP), random forest (RF), Extra-tree classifier (ETC) and decision tree (DT) were employed for classification of data as normal or intrusive. Model performance was analyzed on four feature subsets extracted from the NSL-KDD dataset. As the effectiveness of an IDS is determined by dimensions of the data, therefore, data was pre-processing to remove unrelated attributes from the dataset. Empirical results reveal that the performance of RF, ETC, and DT was above 99% for all attack classes using different feature subsets. So, the proposed model has an effective prediction rate, and also reduces the computational complexity by removing irrelevant features.

Keywords— *SVM; IDS; KNN; Machine learning; Probe; DoS*

I. INTRODUCTION

As cyber-attacks are escalating progressively, security has become a prime concern in computer networks. Moreover, with the advent of technologies like the cloud, networks are being exposed to various intrusive activities consequently, leading to severe impairments. Also, they can cause immense monetary damages and have an unconstructive impact on crucial IT framework thus, causing data inadequacy in cyberwar. So, it is imperative to incorporate security mechanisms to guarantee that the security of the system is not held back. The objective of a secure system is to identify anomalies so that insightful information can be protected. Even though safety mechanisms like firewalls and encryption techniques have been deployed in spite of these, several attacks bypass the security measures of the system. So, it is essential to recognize them at the earliest to decrease the prospective damage to the critical resources, and accordingly, suitable actions can be taken into consideration to get rid of intrusion. An intrusion detection system (IDS) being robust, is the most powerful tool to detect undesirable attempts to

access, manipulate, or disable a computer system, mainly through the internet. It monitors incoming and outgoing traffic to detect malicious activities that jeopardize the security of the system. So, it is an essential tool for network administrators as in absence of such a device, it is infeasible to scrutinize a colossal amount of data traversing the network every second. IDS can be generally be categorized as a misuse intrusion detection system (MIDS) and anomaly intrusion detection system (AIDS) [1]. In misuse detection, the user's activity is compared with known attack signatures, and accordingly, if there is a match the corresponding action is labeled as an attack whereas in the latter any variation from normal behavior is considered as an attack [2]. AIDS is incapable of detecting new attack types. Nevertheless, they have less detection rate as compared to MIDS. Furthermore, an adequate dataset must be available to evaluate the performance of IDS so that the performance of the system can be analyzed before deploying it in the real world. Thus, researchers use a dataset based on which the model is trained and tested. However, it remains a challenge as only limited datasets are publically accessible, and a few among them even lack comprehensiveness and completeness. Some of the commonly used datasets for intrusion detection are KDD-99, NSL-KDD, UNSW-15, Botnet. The primary challenge of IDS includes the problem of misjudgment, false detection, and lack of real-time response. Currently, due to the power of computing machines, technologies like machine learning (ML) have gone under tremendous developments. Subsequently, as ML classifiers drastically boost the accuracy and robustness of the system, they are being applied to the security domain to report various attacks. In recent times, researchers have employed different ML approaches like support vector machine (SVM) [3],[4], decision tree (DT) [5],[6], multi-layer Perceptron (MLP) [7], K-Nearest neighbor (KNN) [8],[9], random forest (RF) [10],[11] for feature selection and classification of network data.

II. LITERATURE SURVEY

An incredible amount of work has been conducted in the field of IDS which is used to identify various attacks on the network and an inclusive review of the same has been presented. ML approaches are comprehensively used to design IDS as they have the ability to handle a massive quantity of data, and accordingly, network administrators can consider adequate measures to prevent the same. Feature selection is indispensable as it removes irrelevant/ redundant features from

the dataset so that the overall competence of the model can be improved. In [12], the wrapper based feature selection method was proposed which makes use of pigeon inspired optimizer to choose the significant features from the dataset. The proposed model was analyzed on KDDCUP 99, NLS-KDD, and UNSW-NB15. A relative study of the model with other existing feature selection approaches was conducted based on which it was concluded that the model performance was better in terms of accuracy. Similarly, in [13], the RF classifier was employed to eliminate unrelated attributes from the dataset so that computational complexity can be reduced. Accordingly, several ML classifiers, namely KNN, SVM, logistic regression (LR), Naïve Bayes (NB), and DT were used to train/test the model. Particle swarm optimization was employed for optimizing model performance and an accuracy of around 99% was obtained on the NSL-KDD dataset when ten significant features were taken into consideration.

SVM has several applications in intrusion detection, [14] used SVM for intrusion detection and the performance was analyzed on the UNSW dataset. Consequently, a relative study with other classifiers such as RF, RepTree, and MLP was conducted yielding a model accuracy of 94%. Likewise, three modified versions of SVM were used by [15] where 99.86% accuracy was obtained in case of improved whale optimized SVM. In [16], various ML classifiers were used to train/ test the model where the Wireshark tool was employed to capture network data packets. The data was initially subjected to pre-processing and an accuracy of 83.6%, 98.2%, 99.8%, 95.1% was attained using NB, SVM, RF, and KNN classifiers. A hybrid approach of knowledge-based and ML was used to detect various types of attacks on KDD-99 [17]. In the proposed system, a knowledge-based approach was used to navigate through the target classes to choose the suitable model based on which prediction was conducted, and accordingly, promising results were attained. In [18], three classifiers, namely MLP, SVM, and NB were analyzed using diverse feature sets. From the experimental results, MLP outperformed the other classifier in detecting mixed, malicious windows and normal traffic. Moreover, the proposed model was fast and thereby, could be effectively used to signify warnings to sophisticated IDS and when it was integrated with traditional IDS, accuracy, and FAR of 92.09% and 0.27% was achieved, respectively. Likewise, [19] used MLP and Neural Networks for identifying network intrusion where the system was initially subjected to training depending on which the data was classified as normal or attack. Moreover, in MLP, the backpropagation algorithm was employed during the testing phase and the model was evaluated on the KDD-99 dataset. When two hidden layers, with and without pre-processing were used in the model, an accuracy of 94% and 91% was attained using all features.

In [20], several feature selection and ranking techniques were taken into consideration for network intrusion detection. Accordingly, the model was trained/ tested on NSL-KDD using an RF classifier. Promising results were obtained using the wrapper method for DoS and probe attacks (41 features)

and U2R on 11 features. Similarly, in [21], random forest and principal component analysis were employed for intrusion detection where the former was used for dimensional reduction, and the latter was used for classification of data. A comparative study with other classifiers such as SVM, NB, and DT was conducted and an accuracy of 96.78% was attained. In [22], IDS was designed using decision tree-based algorithms and the performance was analyzed on CICIDS and BOT-IoT datasets where the accuracy of 96.665% and 96.995% was attained, respectively. Likewise, in [23], a DT classifier with temporal and fuzzy rules was used to train/test the model on the KDD-99 dataset where dynamic recursive feature selection algorithm was used for feature selection. An accuracy of 99.99%, 92.67%, 95.23%, and 57.39% was attained for DoS, Probe, R2L, and U2R respectively.

III. METHODOLOGY

The result of the proposed model has been discussed in this section where SVM, KNN, LR, NB, MLP, RF, extra tree classifier (ETC), and DT were used to label the data as normal or intrusive. The model performance was examined on four different feature subsets that were extracted from the NSL-KDD dataset.

Subsequent steps summarize the process and are precisely discussed in the following sections. In the first step, the data is pre-processed. The second step is to identify the four substantial feature subsets from the model evaluated. The third step is to employ several Machine Learning classifiers for training/ testing the data. And finally, the results of the various parameters are analyzed.

Fig. 1. System Framework

A. Data set and Preprocessing

In the current work, NSL-KDD was employed for training/testing the model, and accordingly, the results were analyzed. During the training phase, 57883, 40047, 10431, 2836 189 instances were used and the test dataset comprises of 19170, 13337, 3646, 913, 63 entries of Normal, DOS, Probe, R2L and U2R extracted randomly from the dataset. The distribution of these records is shown in Fig.2.

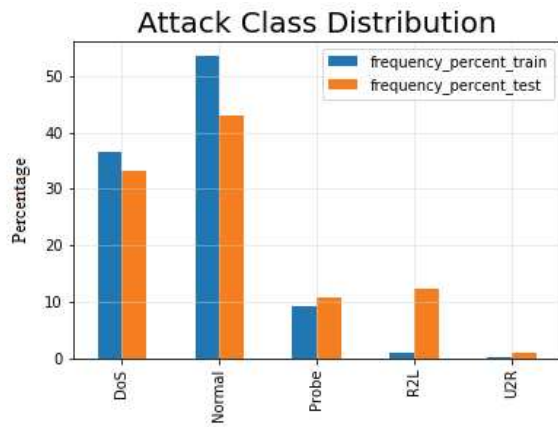


Fig. 2. Attack class distribution

Preprocessing is an essential step in which symbolic or non-numeric attributes are replaced/ removed. To reduce the computational intricacy of the model and simultaneously boost its accuracy, in the current work, four feature subsets presented in Table I was taken into consideration. Since only pertinent attributes were used to train/ test the model, training time was significantly reduced. Thus, pre-processing of data is imperative for dimensional reduction thereby, minimizing the overhead of processing the data.

Table I. DIFFERENT FEATURES SUBSETS EXTRACTED FROM NSL-KDD DATASET

Feature Subset	Selected Attributes
Attribute Set 1	2-8,10,12,13,22-41
Attribute Set 2	1,2,3,5,6,10,12,23,24,25,26-41
Attribute Set 3	1-6,9-19,22-28,31-41
Attribute Set 4	1-6,10-14,16-19,23-25,27-30,32-37,39-41

B. Classification

In the current work, SVM, KNN, LR, NB, MLP, RF, ETC, and DT were used as an intrusion analysis engine to classify the data as normal or intrusive. SVM is a binary-class technique which is applicable for linear and non-linear data. It first maps the input vector, and then an optimal hyper-plane is obtained which is the one that divides the data based on the largest separation between the support vectors. Instead of using the whole training samples, only the support vectors are used by SVM to determine the hyper-plane, and thus, it is exceptionally robust to outliers [24]. In the current work, radial basis kernel function was used to train/ train as using this function promising results are obtained in the case of non-linear data. Similar to SVM, KNN is a supervised learning classifier that uses Euclidean distance and the value of K to classify the data into various classes. Here, K was set to 5 based on which model performance was evaluated thereby, determining the proper position of data points. This process is repeated until convergence, and consequently, the data is classified into various classes [25]. LR is a supervised learning scheme that is used to calculate the possibility of a target variable. It is based on predictive analysis and describes the association among dependent and independent variables. In the current work, uniform weights were associated with all the

classes to obtain unbiased results and the value of maximum iterations was set to 100 so that the model could learn in an efficient manner. Naive Bayes, a collection of classifiers uses Bayes theorem which follows the scheme that every record can independently be classified. In the current work, Gaussian NB was used as it works well for numeric data. MLP is a neural network comprising of nodes or neuron which map the input values to the subsequent output values. It has a single input/ output layer, and numerous hidden layers to classify linear or non-linear data [26]. The model was trained/tested using a rectified linear unit function where 100 neurons present on a single hidden layer were employed.

In a decision tree, a tree-like structure is formed to solve the classification problem. The nodes of the tree denote the attributes whereas the paths indicate the possible values of the attributes [27]. Nodes are split using information gain or Gini index criteria (GI) and in the current work, GI was used to choose the best node for the split. DT uses a recursive process which stops when all the data has been classified as the depth of the tree was set to none. The random forest takes into account the output of various decision trees to classify the data into appropriate classes. In the case of RF, significant attributes are selected from a random set of attributes which results in the formation of a better classification model [28]. The extra-tree classifier is similar to RF, and the only difference lies in the process of tree construction. It combines the results of numerous de-correlated decision trees in a forest to classify the data and can be effectively used for network intrusion detection. In the case of RF and ETC, the number of trees was set to 50, 100, respectively to determine the best node for the split.

IV. RESULTS

In current work, features from NSL-KDD were selected randomly to shrink the dimension of data based on which model was trained/ tested. Random selection of features is an effective way to reduce training time and model intricacy. By analyzing the results, this technique worked well in the current model although it might be unfavorable in some scenarios. Here several classifiers namely, SVM, KNN, LR, NB, MLP, RF, ETC, and DT were trained/ tested on 111386 and 37129 instances of NSL-KDD and the model performance was evaluate on different parameters. A graphical representation of results in form of accuracy, precision, recall, and F1-score on various feature subsets is depicted in figures below.

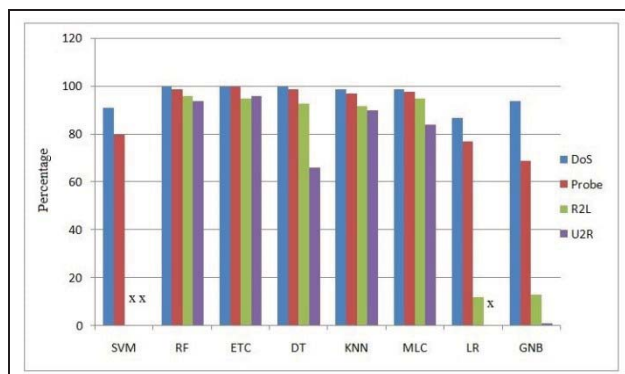


Fig. 3.a. Attribute set 1

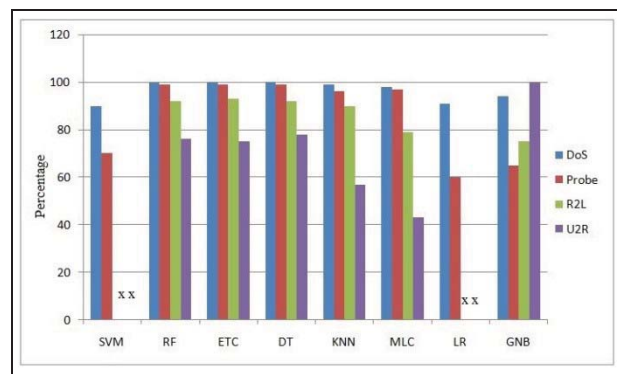


Fig. 4.a. Attribute set 1

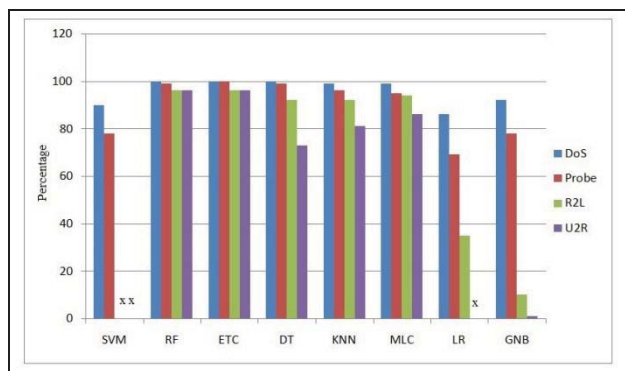


Fig. 3.b. Attribute set 2

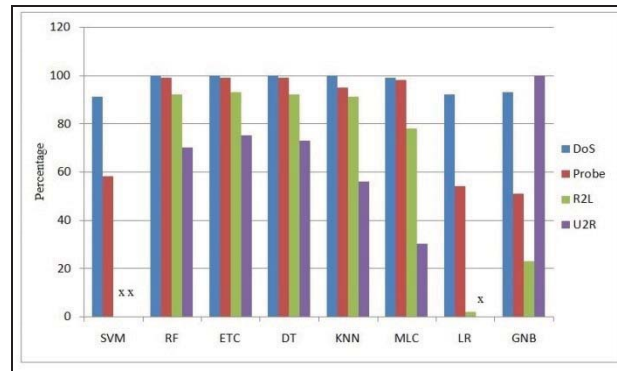


Fig. 4.b. Attribute set 2

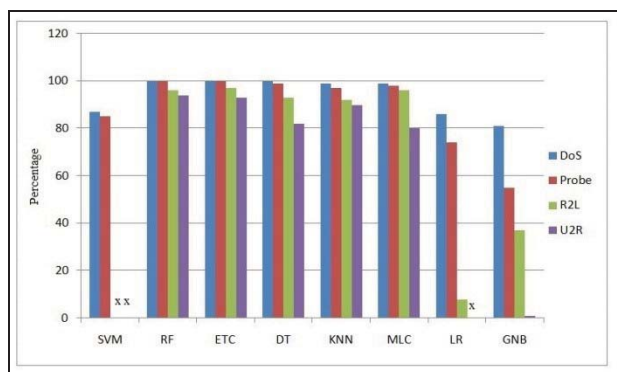


Fig. 3.c. Attribute set 3

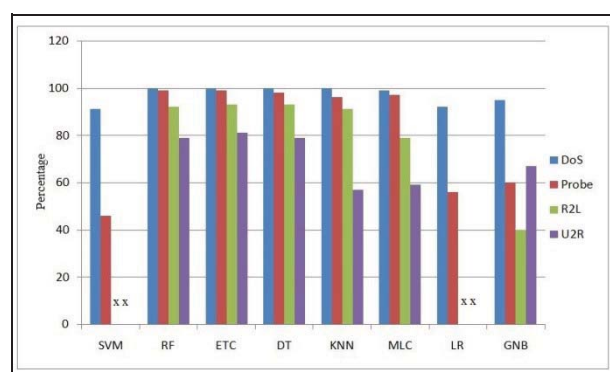


Fig. 4.c. Attribute set 3

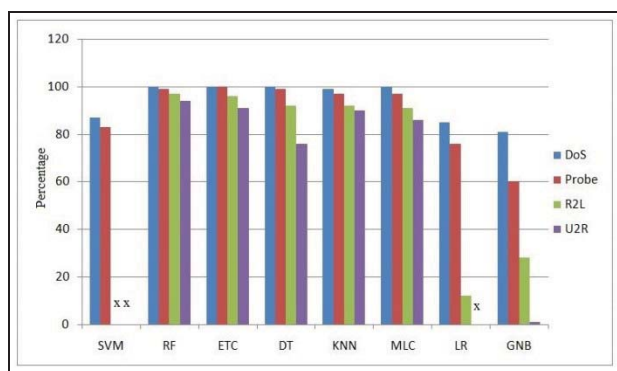


Fig. 3.d. Attribute set 4

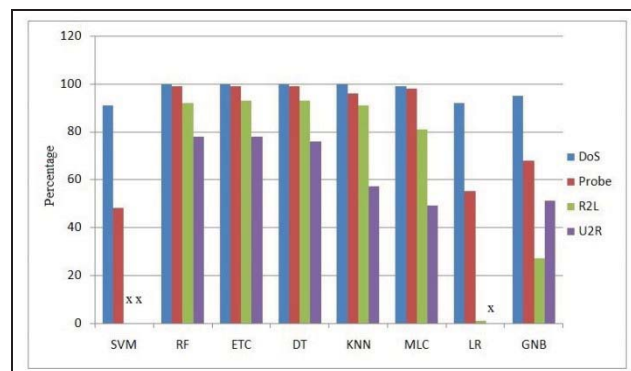


Fig. 4.d. Attribute set 4

Fig. 3. Graphical representation of Precision using various feature subsets

Fig. 4. Graphical representation of recall using various feature subsets

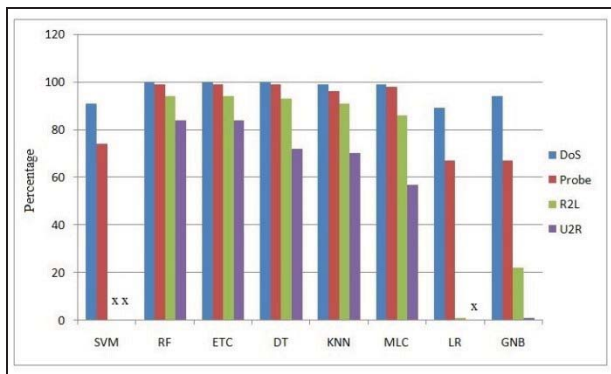


Fig. 5.a. Attribute set 1

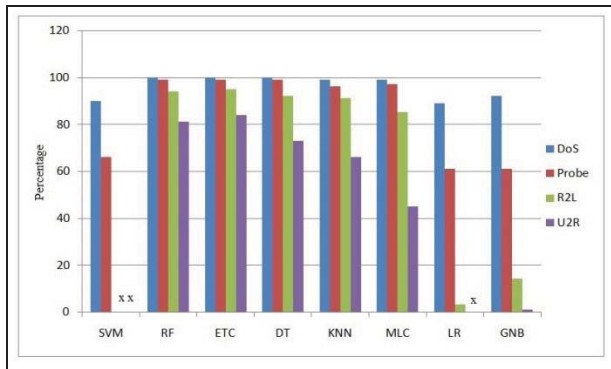


Fig. 5.b. Attribute set 2

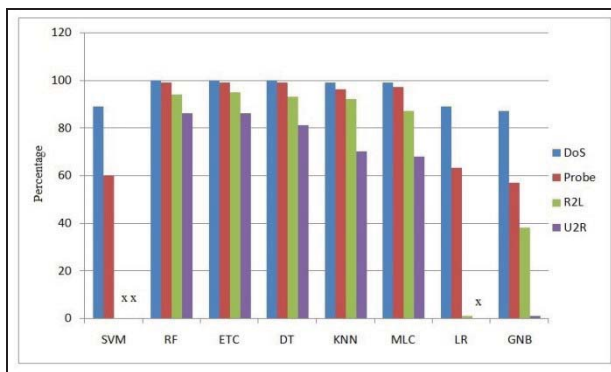


Fig. 5.c. Attribute set 3

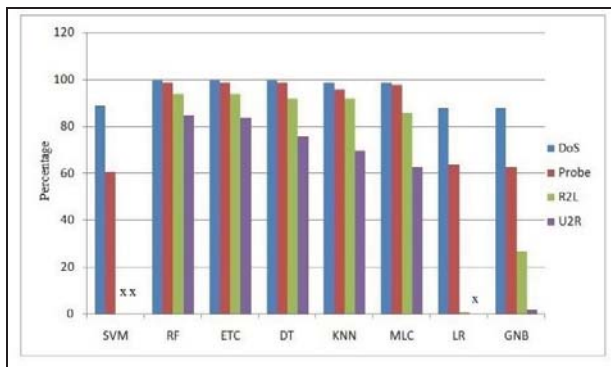


Fig. 5.d. Attribute set 4

Fig. 5. Graphical representation of F1- score using various feature subsets

Table II. MODEL ACCURACY OF VARIOUS CLASSIFIERS

Expert	Attribute set 1	Attribute set 2	Attribute set 3	Attribute set 4
SVM	89.31%	88.07%	87.45%	87.65%
RF	99.47%	99.46%	99.48%	99.48%
ETC	99.51%	99.53%	99.54%	99.52%
DT	99.18%	99.21%	99.21%	99.21%
KNN	98.57%	98.47%	98.62%	98.64%
MLP	98.03%	98.00%	98.29%	98.44%
LR	87.05%	86.03%	86.64%	86.66%
NB	54.11%	61.68%	63.34%	73.93%

Among the attack classes, promising results were obtained in case of DoS attack as there are reasonable entries of this attack class in the NSL-KDD dataset whereas poor results were attained for the U2L attack. Overall, the accuracy of above 99% was achieved using random forest, extra-tree, and decision tree classifiers on all four feature subsets, as they can handle unevenly dispersed data. A random forest can recognize data imbalance and it uses the bootstrap process to increase the minority class occurrence thereby, reducing misclassification of data, thus, leading to a rise in the accuracy. Likewise, DT has excellent generalization ability and performs inclusive scrutiny of all possibilities along its branches resulting in an exceptional performance. Also, these classifiers aim to limit the cost thereby, making them better predictors for intrusion detection. Thus, they can be effectively used for network intrusion detection.

V. CONCLUSION AND FUTURE SCOPE

Security has become a severe concern owing to increased unauthorized access and exploitation of network resources. As attacks have become a detrimental problem, it is important to detect them promptly so that the damage to the system network can be minimized. Recently, machine learning classifiers have been extensively employed in IDS, due to their adaptability, generalization capability, and robust nature. In current research work, inclusive empirical study using ML classifiers, namely SVM, KNN, LR, NB, MLP, RF, ETC, and DT was conducted to detect network intrusion and the performance was evaluated on NSL-KDD. The dataset was initially pre-processed and depending on the significant attributes the model was trained/ tested. The empirical results reveal that all the classifiers generate promising results for Dos attack, whereas for U2R least accurate results were obtained owing to their instance in the NSL-KDD dataset.

For future, the research work can be focused on the prospect of applying optimization techniques so that better

IDS can be offered. Consequently, ensemble-based approaches can be examined, which combines the output of various algorithms to predict the final results. Besides, diverse feature selection methods can be analyzed to lessen computational intricacy of the system. Furthermore, the problem of intrusion detection needs to be addressed in wireless mobile network domain.

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